



សាកលវិទ្យាល័យអន្តរជាតិ
PHNOM PENH INTERNATIONAL UNIVERSITY

UrbanCart

Big Data Analytics

Subject: Big Data

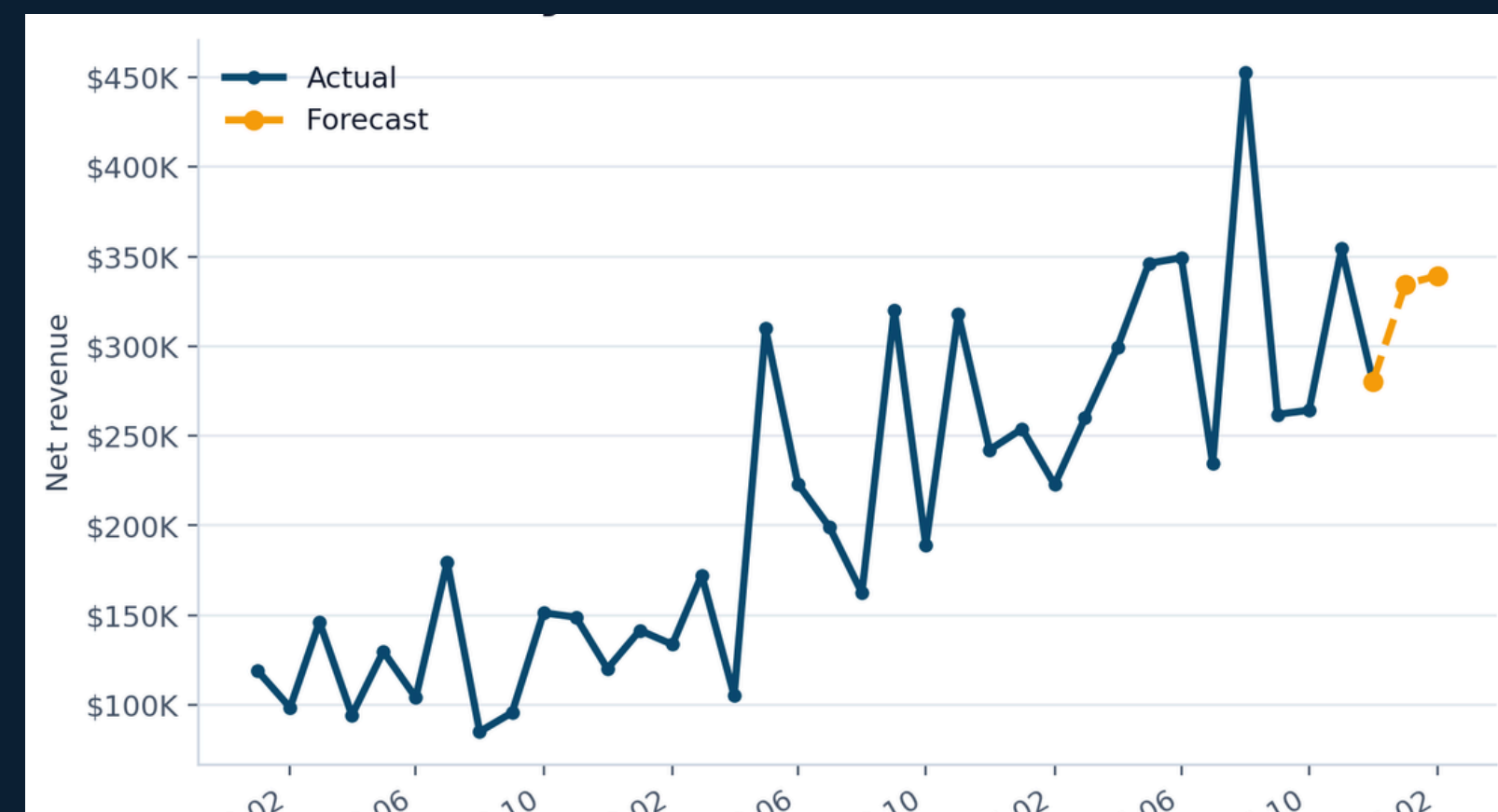
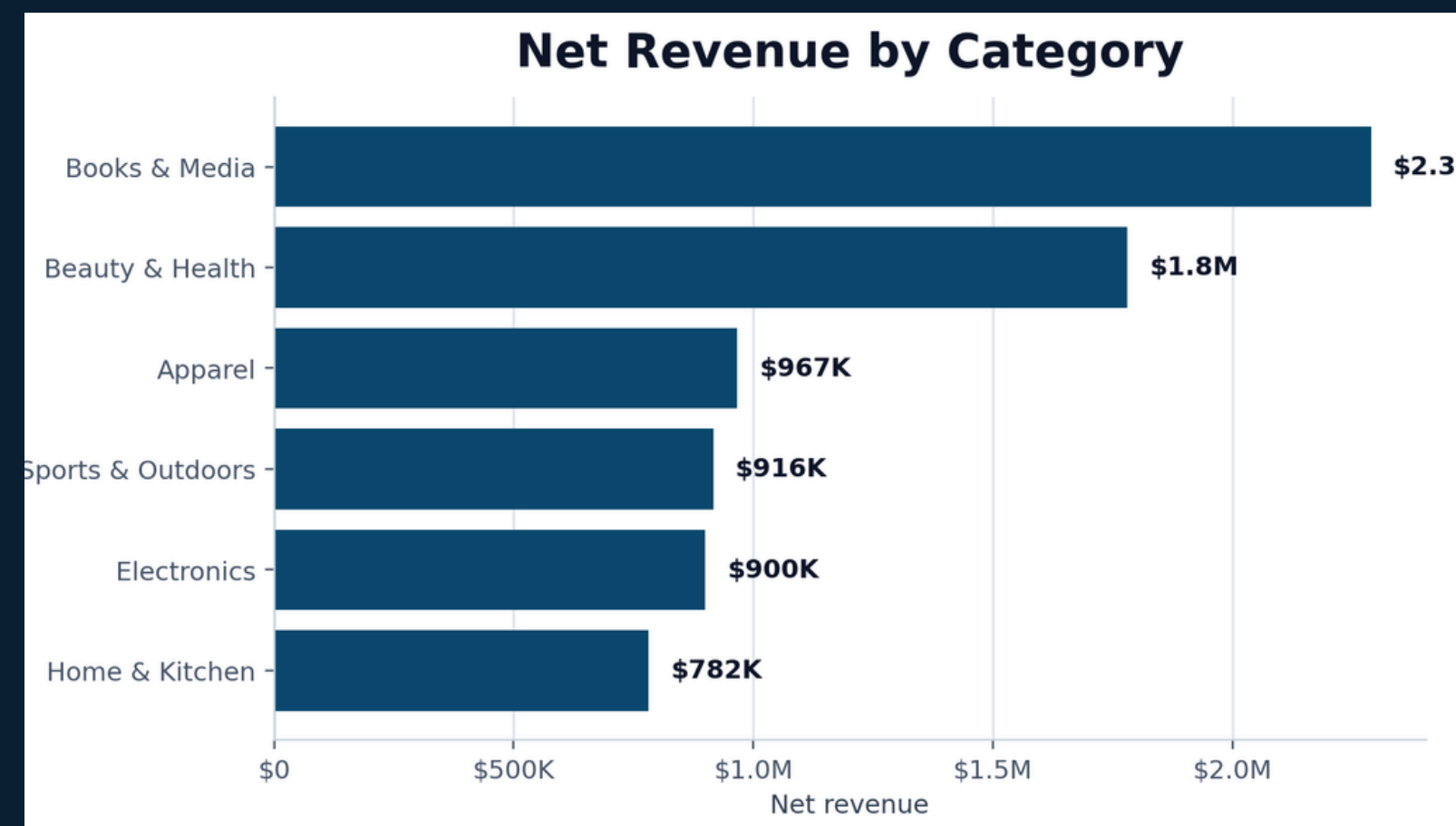
Room: Accounting Lab

Lecturer: Mr. Min Sothearith

Group Members:

Keo Samnang • Sue Theara • Teng Nyka • Yin Panharith

SQL | pandas | NumPy | Matplotlib | Jupyter



Presentation Roadmap

Team members

Keo Samnang	SQL Queries 1–4, database loading, date standardization, and missing-value handling.
Sue Theara	NumPy methods, business charts, report preparation, and reusable pipeline.
Teng Nyka	SQL Queries 5–10, outlier handling, product catalog reconciliation, and time-series analysis.
Yin Panharith	GitHub repository organization, README, requirements, presentation slides, and project validation.

What we will cover

- Business problem
- Data pipeline
- Cleaning evidence
- NumPy methods
- 8 business findings
- Recommendations + deliverables

Business Context

PROBLEM TO SOLVE

Messy ecommerce data makes business decisions risky.

- Who are the most valuable customers?
- Which categories and products really perform?
- Can legacy and supplier data be trusted?

Goal

Raw data → clean data → models → business actions

8

business findings

1

reusable pipeline

Data Sources & Workflow

Data scale

2,500

customers

9,000

orders

20,176

clean order items

12,000

web sessions

Workflow

SQL extraction → pandas cleaning → NumPy methods → charts → pipeline

External files: legacy customer export + supplier product catalog

Output: clean datasets, model results, charts, report and GitHub repository.

Phase 2 — Cleaning Evidence

AUDIT RESULTS

1,377

legacy customer records kept

0

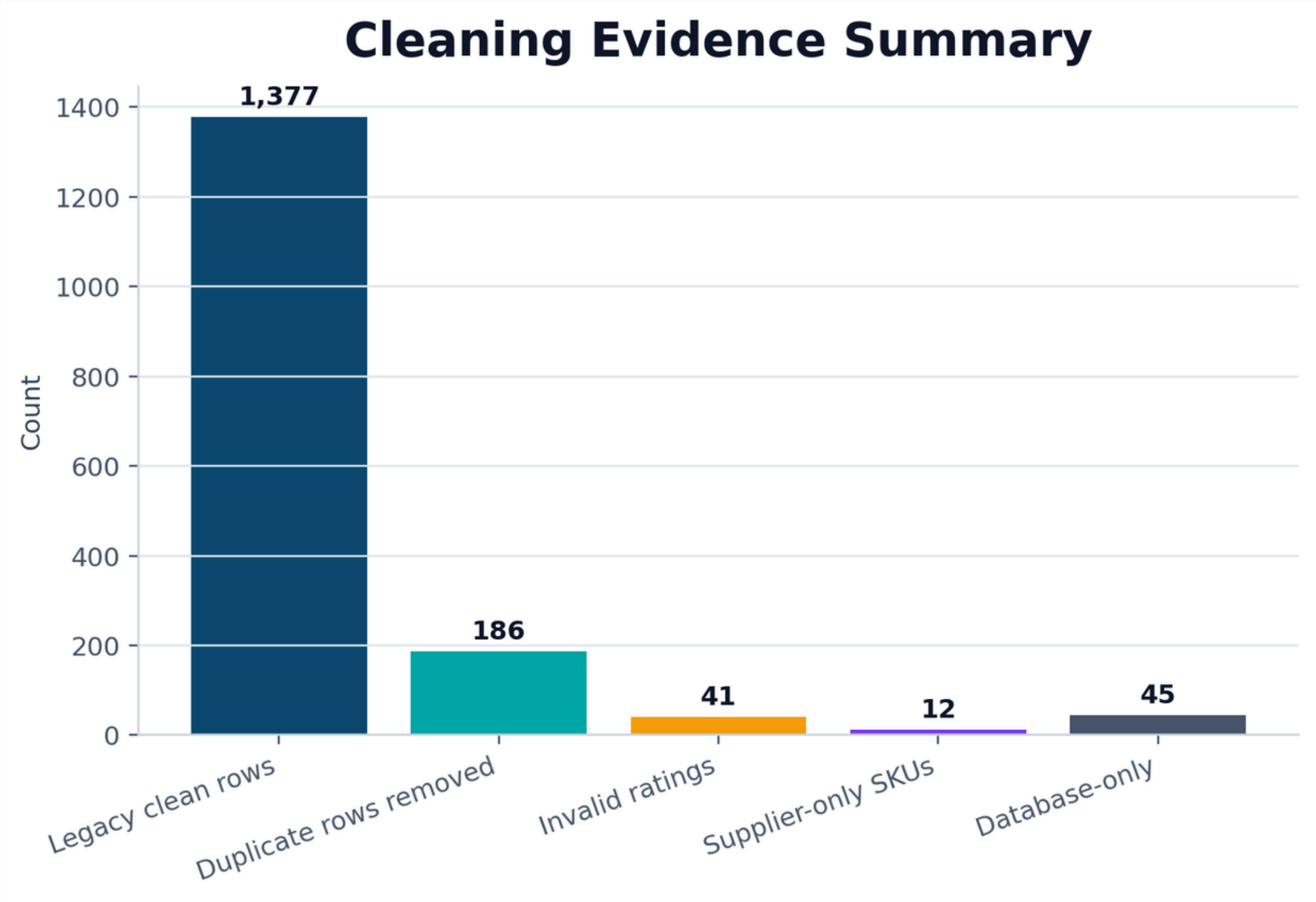
unparsed signup dates

186

duplicate order rows removed

Key policy

Do not silently fill everything with zero. Keep flags and audit tables.



Methods Used

PHASE 1 TO PHASE 5

SQL	10 queries + window functions	10	9
pandas	cleaning + integration	SQL queries	clean datasets
NumPy	RFM, similarity, regression, simulation	4	8
Matplotlib	8 business charts	NumPy methods	insight charts
Pipeline	python main.py		

The next slides are the core findings.

Finding 1 — Category Revenue

BUSINESS INSIGHT 1 OF 8

Which category generates the most revenue?

Books & Media

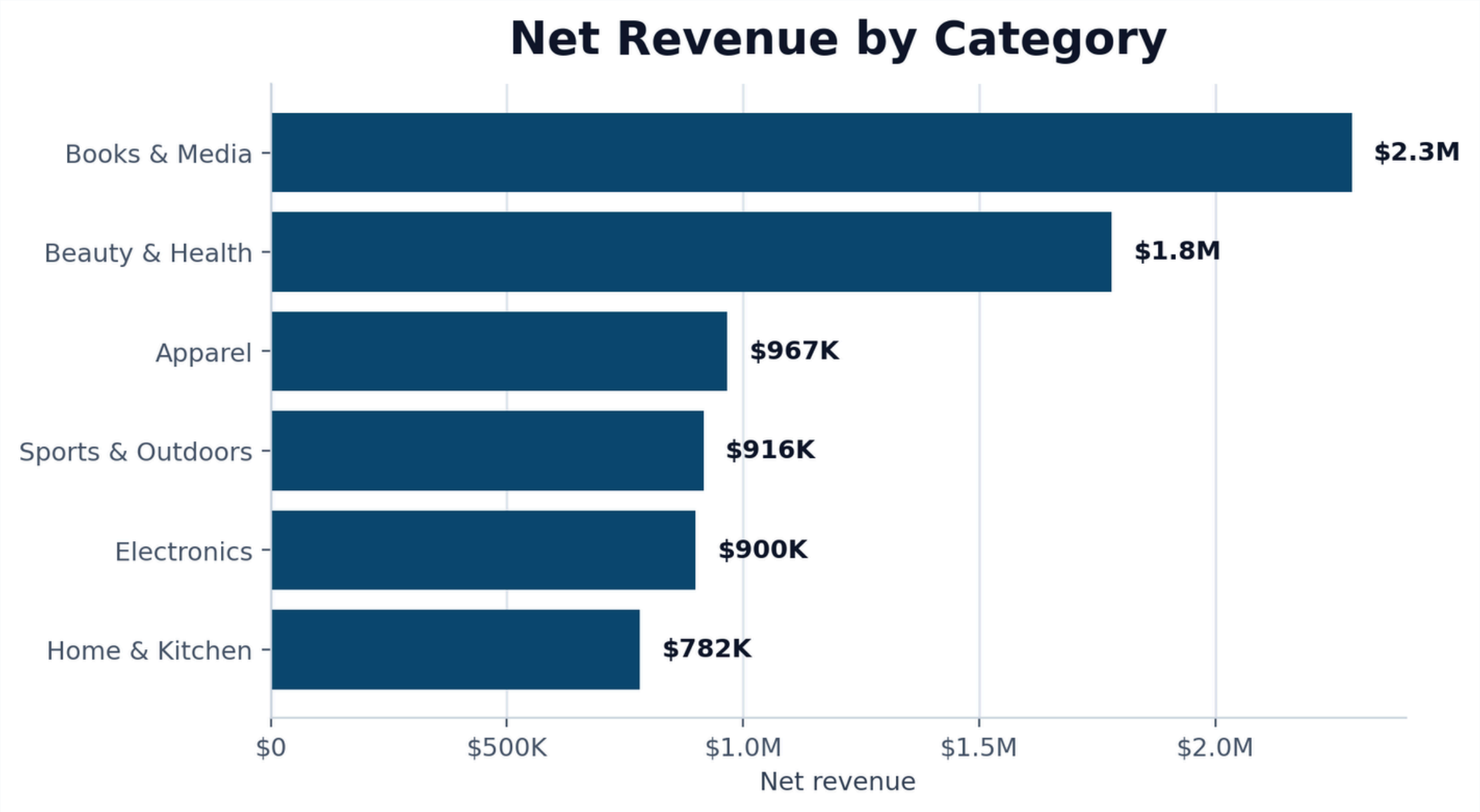
\$2.29M net revenue

Meaning

Books & Media leads the business, with Beauty & Health close behind.

Action

Protect inventory and campaign budget for top categories.



Finding 2 — RFM Customer Value

BUSINESS INSIGHT 2 OF 8

Which segment creates the most value?

Champions

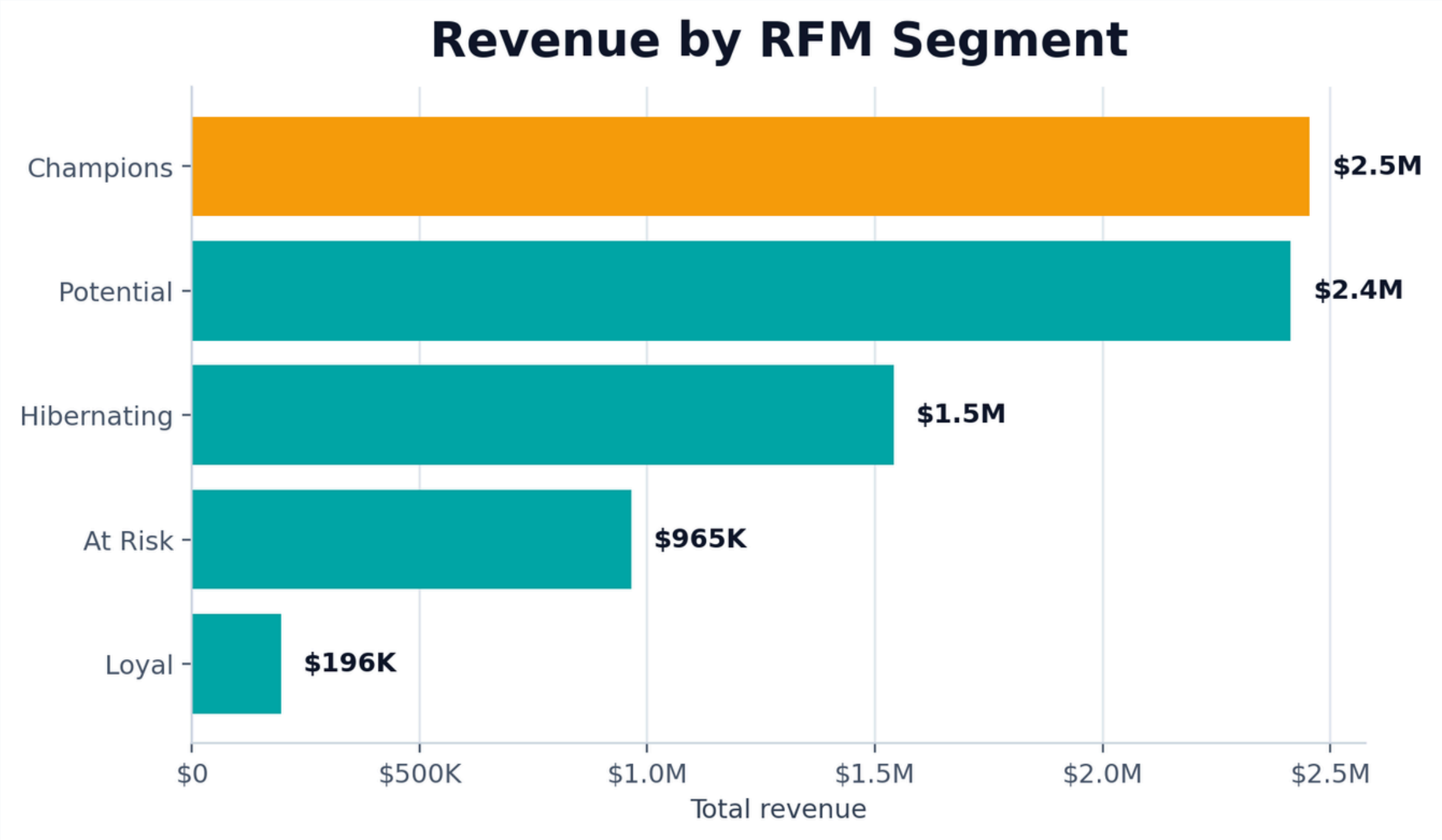
\$2.46M revenue

Meaning

A small customer segment contributes a very large revenue share.

Action

Give Champions loyalty rewards and early churn monitoring.



Finding 3 — Return Rate

BUSINESS INSIGHT 3 OF 8

Where is return risk highest?

2.45%

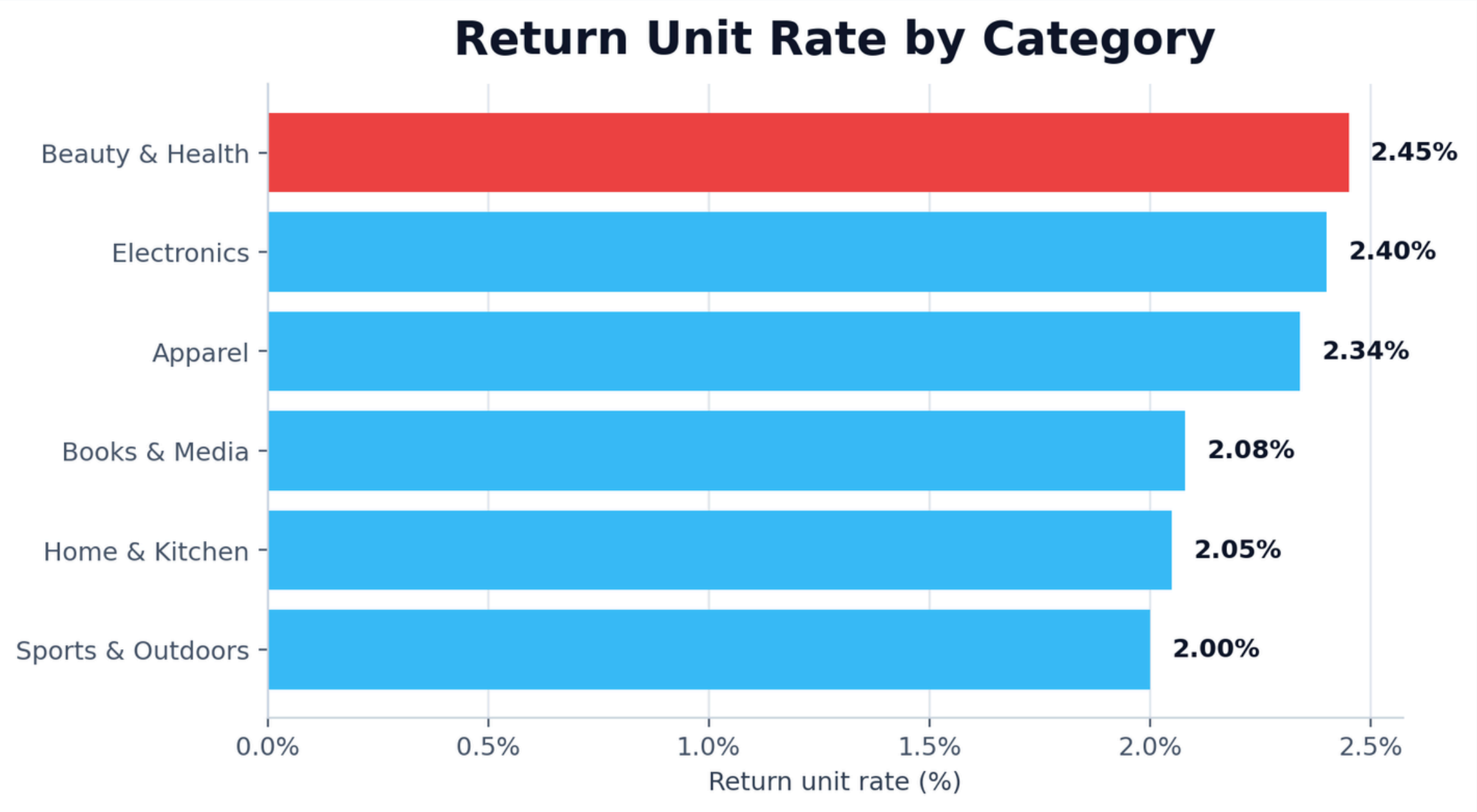
highest return rate

Meaning

Returns can reduce profit and may point to quality or description issues.

Action

Inspect suppliers, product pages and packaging in Beauty & Health.



Finding 4 — Revenue Forecast

BUSINESS INSIGHT 4 OF 8

What is near-term revenue outlook?

\$334.6K

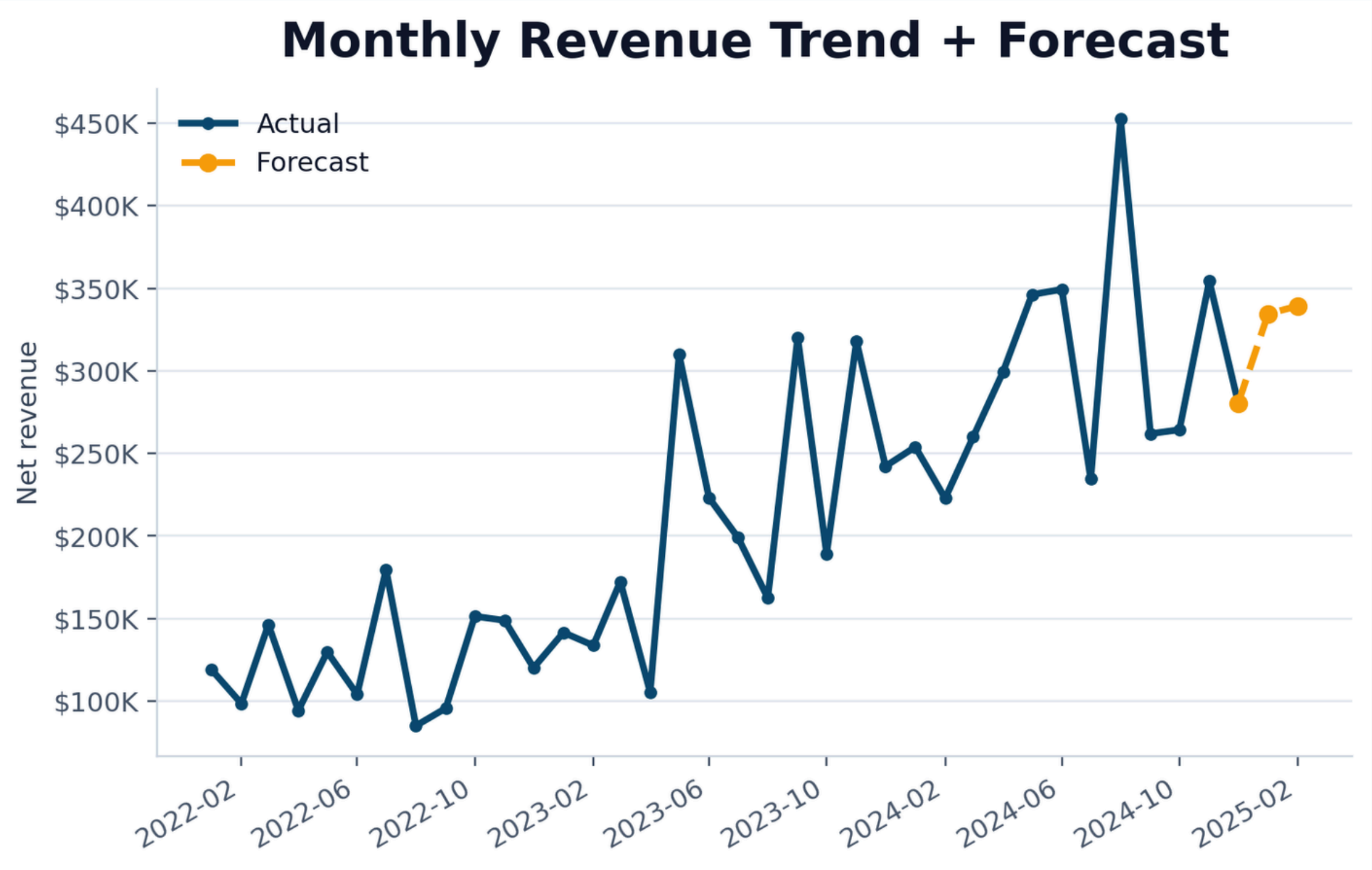
month 1 forecast

Meaning

Revenue is volatile, so the forecast should be used as a planning range.

Action

Refresh the forecast monthly and prepare demand ranges, not fixed targets.



Finding 5 — Top Products

BUSINESS INSIGHT 5 OF 8

Which products need inventory protection?

Top 8

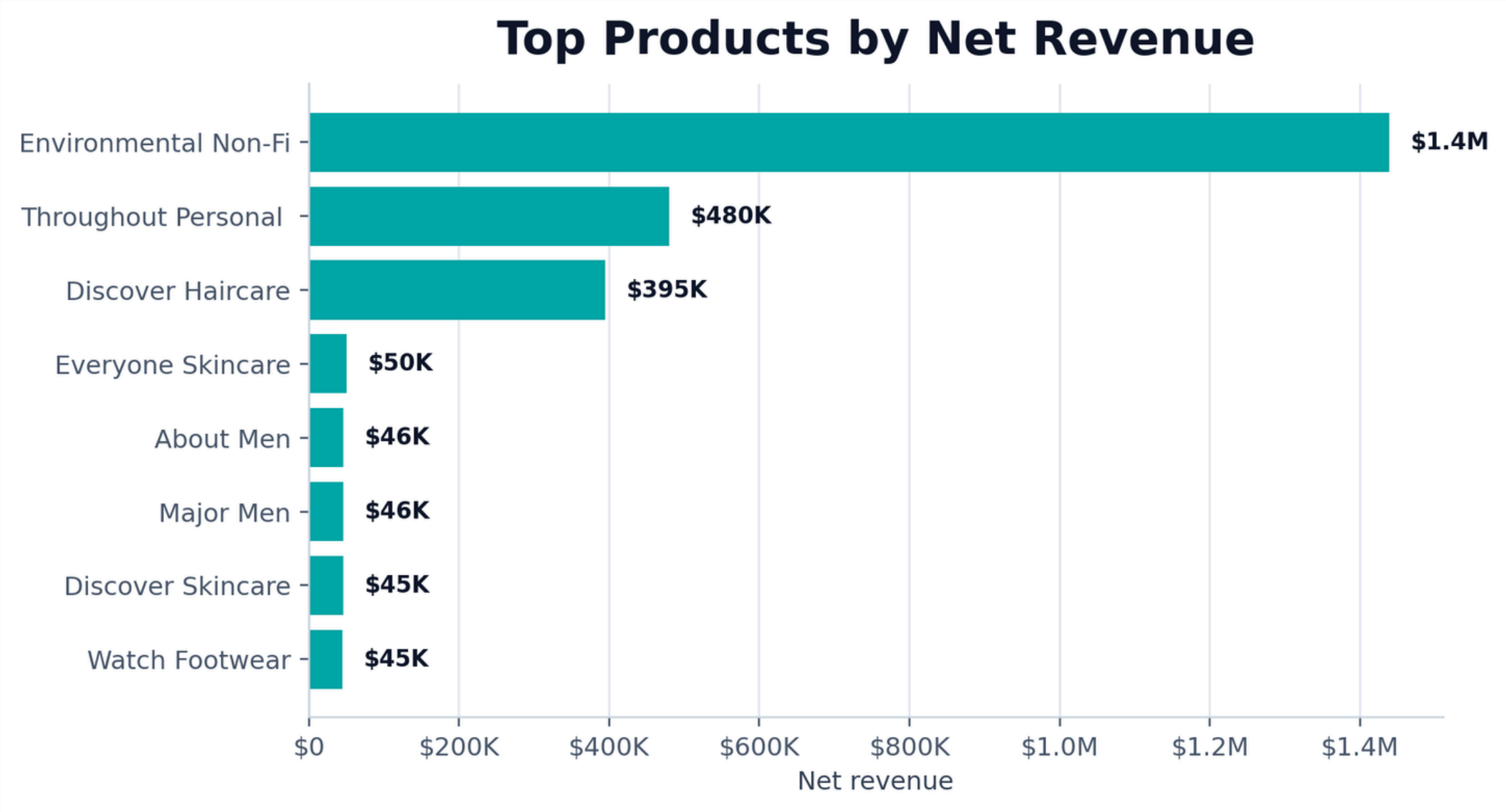
products shown

Meaning

A small set of products drives a large part of net revenue.

Action

Prioritize replenishment and recommendation placement for high-revenue products.



Finding 6 — Payment Mix

BUSINESS INSIGHT 6 OF 8

Do countries use the same payment methods?

22.27%

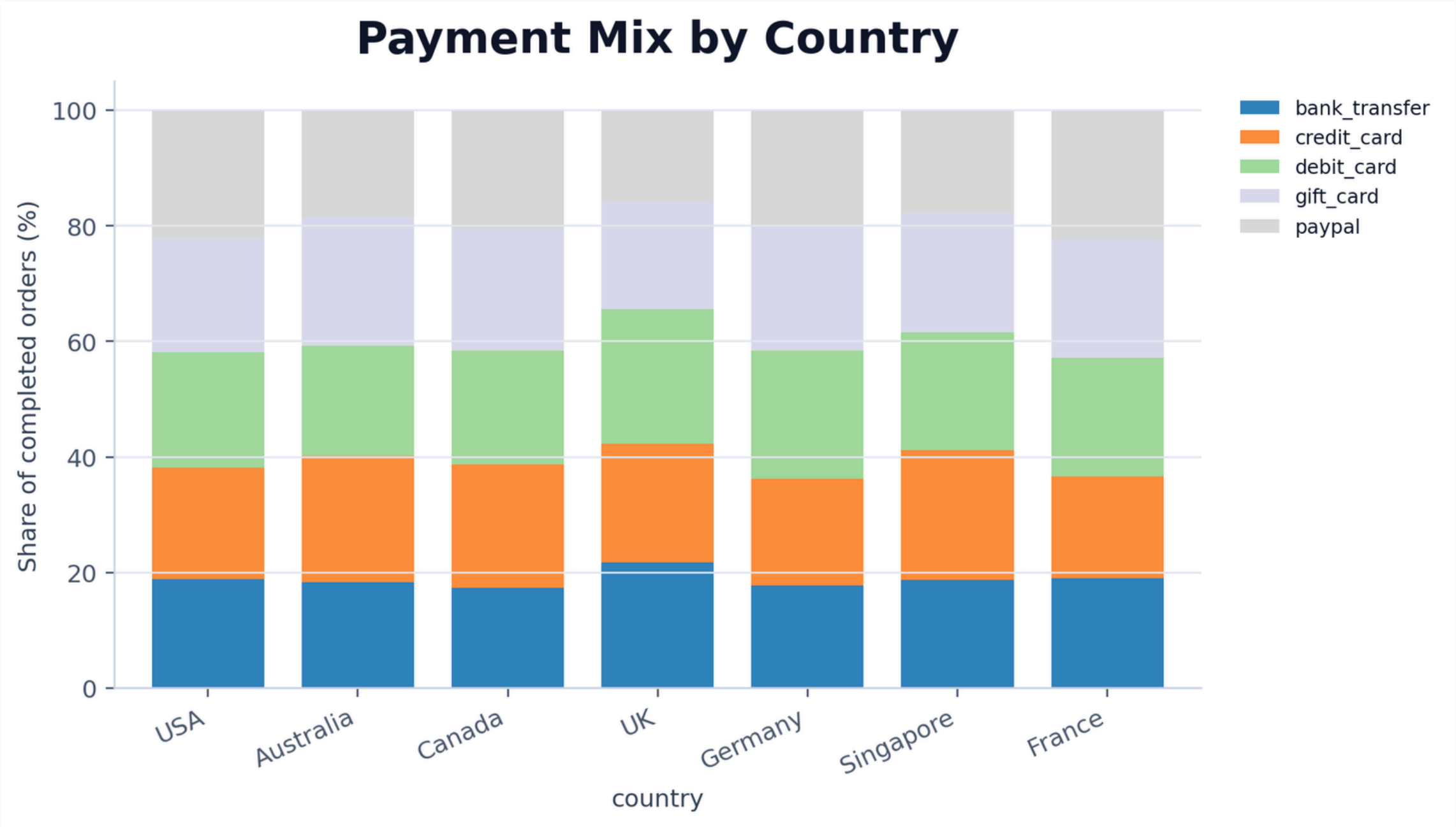
Australia gift cards

Meaning

Checkout design should not assume one universal payment preference.

Action

Localize payment options and test provider ordering by country.



Finding 7 — Ratings vs Repeat Purchase

BUSINESS INSIGHT 7 OF 8

Do ratings explain repeat purchase?

$r = 0.035$

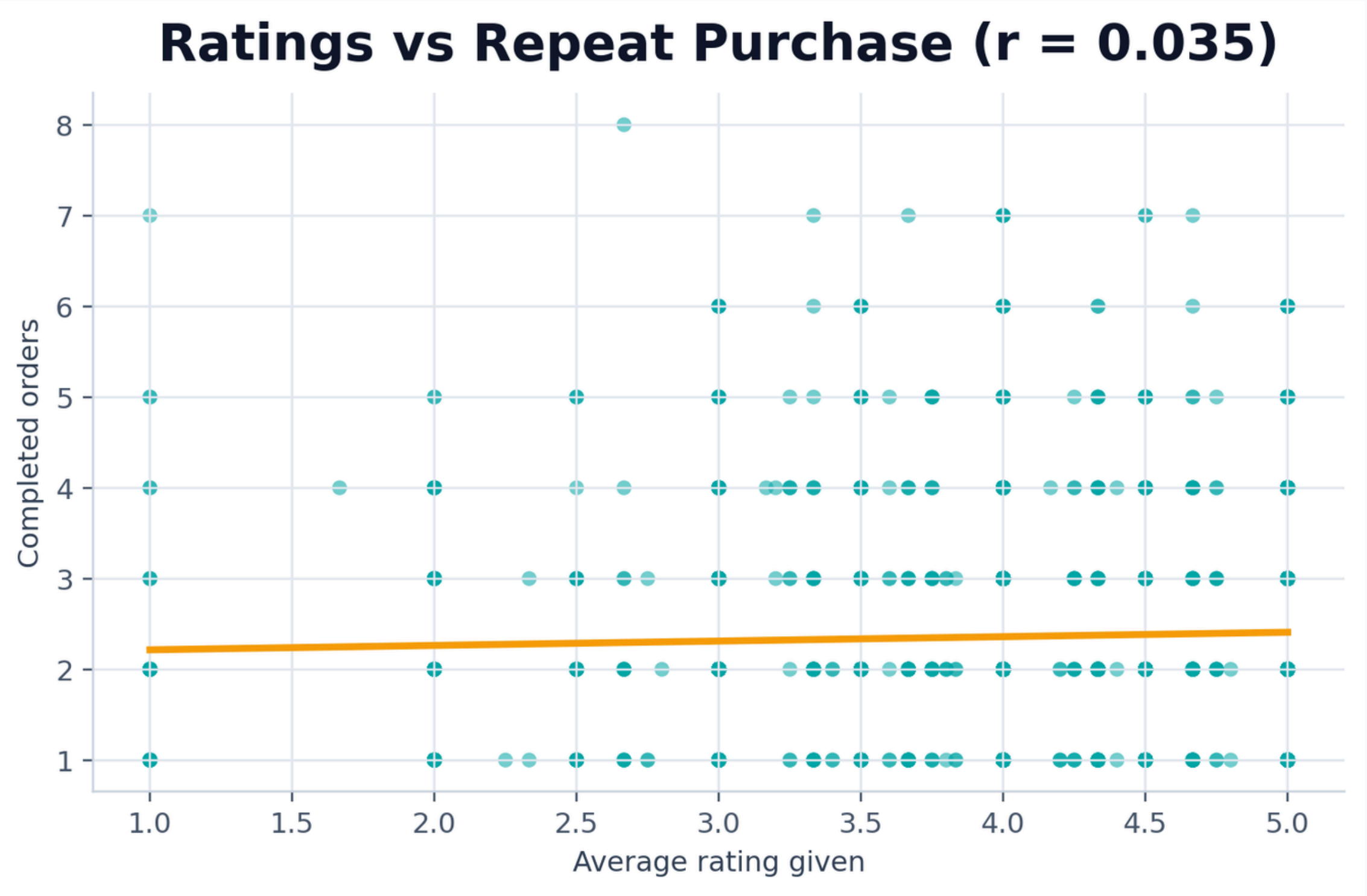
weak relationship

Meaning

Ratings alone are a weak loyalty signal.

Action

Combine ratings with RFM value, recency and web engagement.



Finding 8 — Weekly Activity

BUSINESS INSIGHT 8 OF 8

Do engagement and revenue move together?

4-week

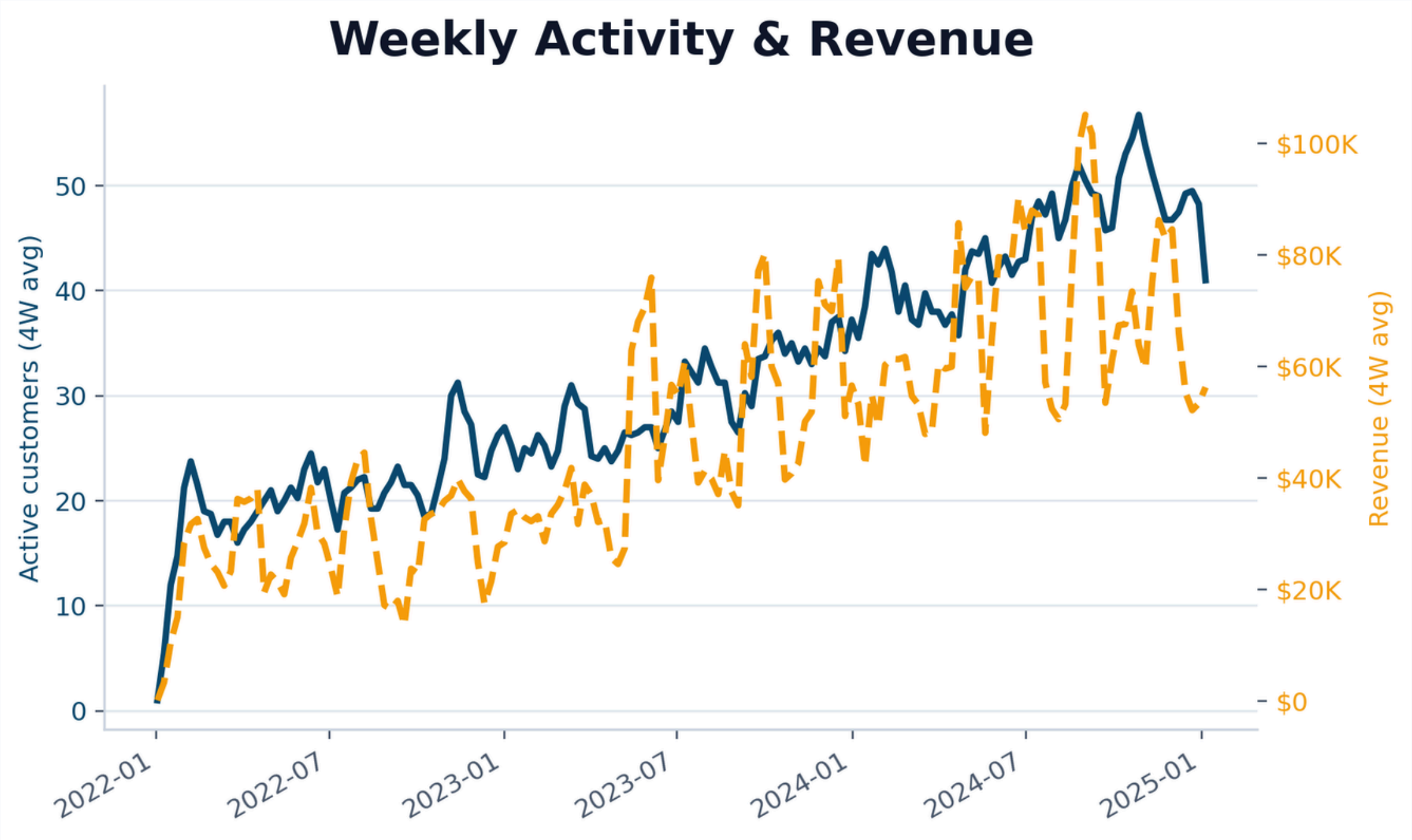
moving averages

Meaning

Active customers and revenue can diverge because of basket size, discounting and product mix.

Action

Monitor both metrics together every week.



Recommendations

- | | | |
|-----------|--|--|
| 01 | Protect high-revenue categories | Keep Books & Media and top products in stock. |
| 02 | Retain Champions | Reward high-value customers and monitor churn risk. |
| 03 | Reduce returns | Investigate Beauty & Health quality and product content. |
| 04 | Improve product governance | Resolve supplier-only and database-only SKU exceptions. |
| 05 | Plan with uncertainty | Use forecasts as ranges and refresh monthly. |

Submission Package

REPOSITORY CONTENTS

queries.sql	10 SQL queries
analysis.ipynb	Phases 1–4 notebook
data/processed	clean reusable datasets
report.pdf	52-page written report
executive_summary.pdf	1-page management summary
README.md	run steps + team contributions
main.py + src/pipeline.py	one-command pipeline

Run command

python main.py

Regenerates clean data, tables, models and charts.

Thank you

Decision-ready analytics starts with clean data.

Group 9

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Q&A

